

KUSHAGRA SHARAN

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EDUCATION

Manipal Institute of Technology

B.Tech in Data Science and Engineering; CGPA: 8.04/10

Manipal, India

Aug 2024 – May 2028

EXPERIENCE

Centre for Railway Information Systems (CRIS)

AI/ML Intern, Government of India PSU under Indian Railways

New Delhi, India

May 2026 – Present

- Engineered TDMS Assistant, a distributed multilingual Text-to-SQL platform (English, Hinglish, Hindi) over structured railway schemas, architecting a 4-service stack (PostgreSQL, ChromaDB, Ollama, Streamlit) containerized with Docker Compose for reproducible, one-command deployment
- Reduced LLM inference latency 12x (approx. 50s to 4s) via context pruning, parameter tuning, and token-budget management on local GPU hardware with no hardware upgrade, directly improving system responsiveness under load
- Fine-tuning a domain-adapted sentence transformer for semantic schema retrieval over ChromaDB and hardening the codebase for open-source release on Hugging Face

Ernst & Young (EY)

AI Intern, Tax Technology and Reconciliation

Gurugram, India

Jul 2025 – Aug 2025

- Shipped 4 production systems in 8 weeks: NLP data pipelines, an LLM-driven financial assistant, a real-time analytics dashboard, and high-volume Selenium automation for B2B invoice retrieval at scale
- Fine-tuned all-MiniLM-L6-v2 via hard-negative mining (MNRL) across 287 schema variants, achieving +9.4% Macro F1 and a 15x latency reduction (0.27ms/column) through batch-encoding optimization
- Built a full-stack conversational analytics assistant (Groq LLM APIs, FastAPI backend, Vite/TypeScript SPA) delivering real-time financial reporting end-to-end

PROJECTS

Multilingual Text-to-SQL RAG System | *Ollama, ChromaDB, PostgreSQL, Docker, Streamlit, FastAPI*

- Built an end-to-end distributed platform supporting English, Hinglish, and Hindi, with the full stack containerized via Docker Compose (PostgreSQL, ChromaDB, Ollama, Streamlit) for one-command reproducible deployment
- Reduced inference latency 12x (approx. 50s to 4s) via context pruning and parameter tuning; built a modular SQLAlchemy ORM layer enabling plug-and-play schema expansion across enterprise databases
- Exposed a RESTful FastAPI interface for downstream system integration and implemented semantic schema retrieval using transformer embeddings over ChromaDB

Semantic GST Schema Standardization Engine | *HuggingFace Transformers, sentence-transformers, Python*

- Built a production Python package mapping noisy financial headers to a 61-field compliance schema via a fine-tuned transformer (MNRL on 287 real-world variants), achieving +8.7% Top-1 accuracy and +9.4% Macro F1
- Achieved a 15x latency reduction via vectorized batch encoding; implemented collision resolution, confidence thresholding, and JSONL audit logging for enterprise-scale throughput

TECHNICAL SKILLS

Languages: Java, Python, C, SQL

Machine Learning & AI: Machine Learning, Deep Learning, NLP, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), LLM Inference Optimization, HuggingFace Transformers, sentence-transformers, scikit-learn

Systems & Infrastructure: Docker, Git, GitHub Actions (CI/CD), Linux, REST APIs, Distributed Systems, SQLAlchemy, Workflow Automation

Databases & Frameworks: PostgreSQL, MySQL, ChromaDB, SQLite, FastAPI, Streamlit, Pandas, NumPy, Selenium

LEADERSHIP AND AWARDS

NASA Space Settlement Design Competition

Head of Human Factors Engineering

Kennedy Space Center, FL

2022 – 2023

- Directed a 50+ member international team, delivering a full orbital habitat proposal to NASA and Boeing engineers
- Received the Dick Edwards Exceptional Leadership Award (top 0.5% of the global cohort); National Winner; Asian Regional and International Runner-Up
- Ranked All India Rank 1 (English) in the Global Talent Search Examination

CERTIFICATIONS

Python for Data Science, AI & Development — IBM | Cambridge English First — CEFR C1, Grade A